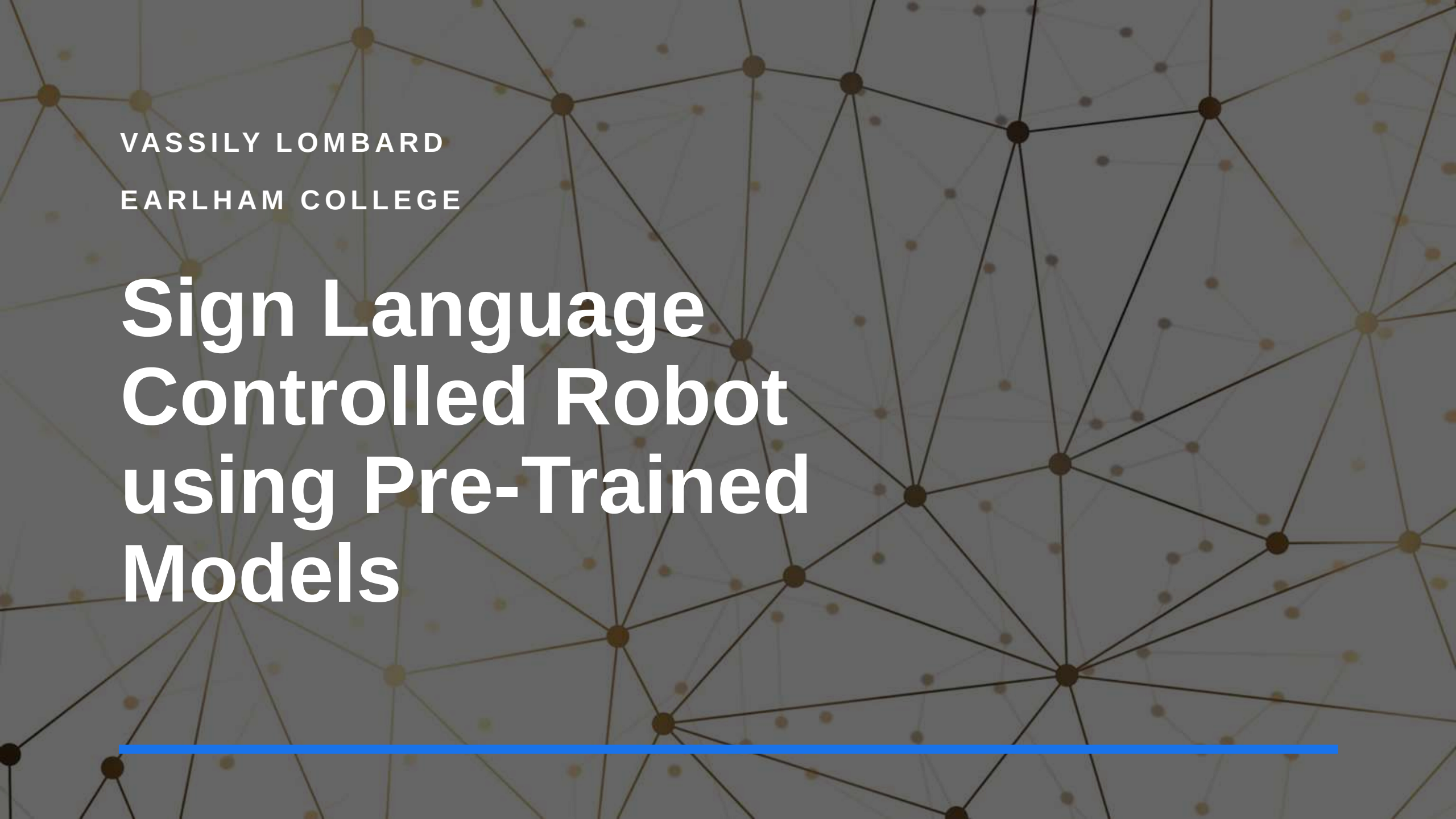
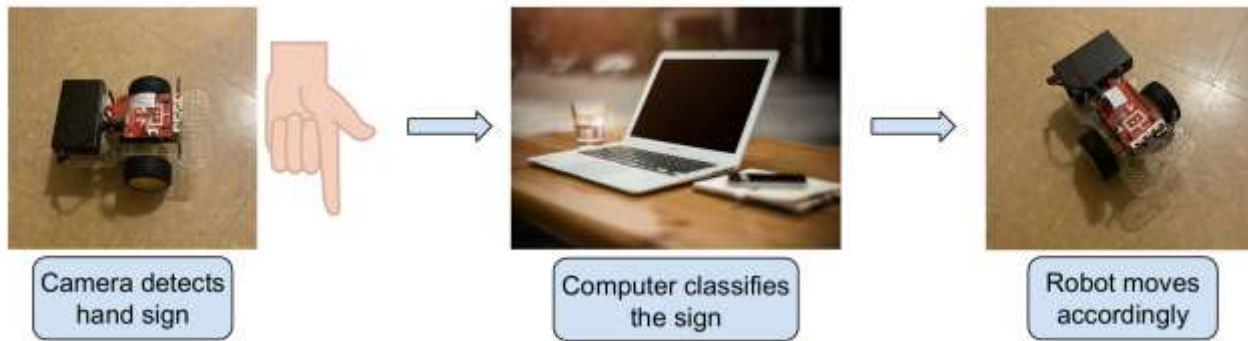


The background of the slide is a dark gray with a complex network of thin, light brown lines connecting various nodes. Some nodes are larger and darker, while others are smaller and lighter. The overall effect is a sense of a global or digital network.

VASSILY LOMBARD
EARLHAM COLLEGE

Sign Language Controlled Robot using Pre-Trained Models

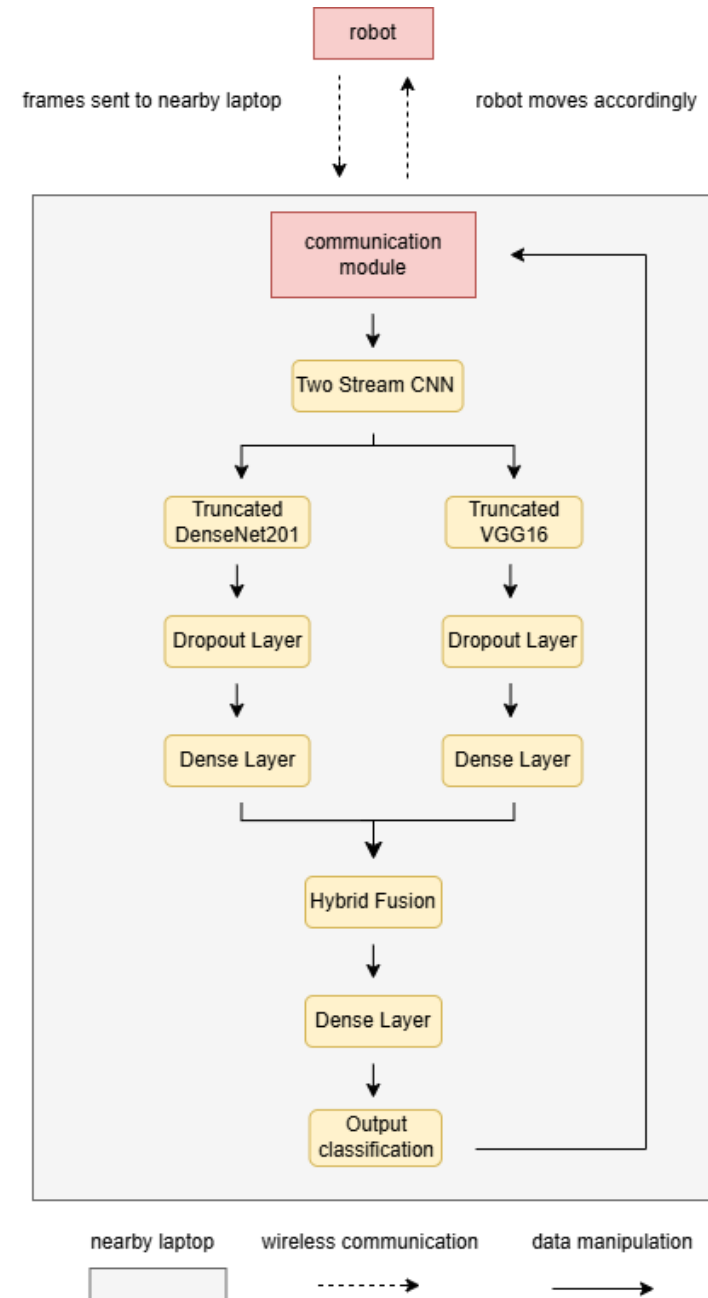
Introduction



- Project goal: build a robot that understands ASL gestures and moves accordingly
 - Uses computer vision pre-trained models and a Raspberry Pi
 - Aims to enhance accessibility and human-centered tech design
-

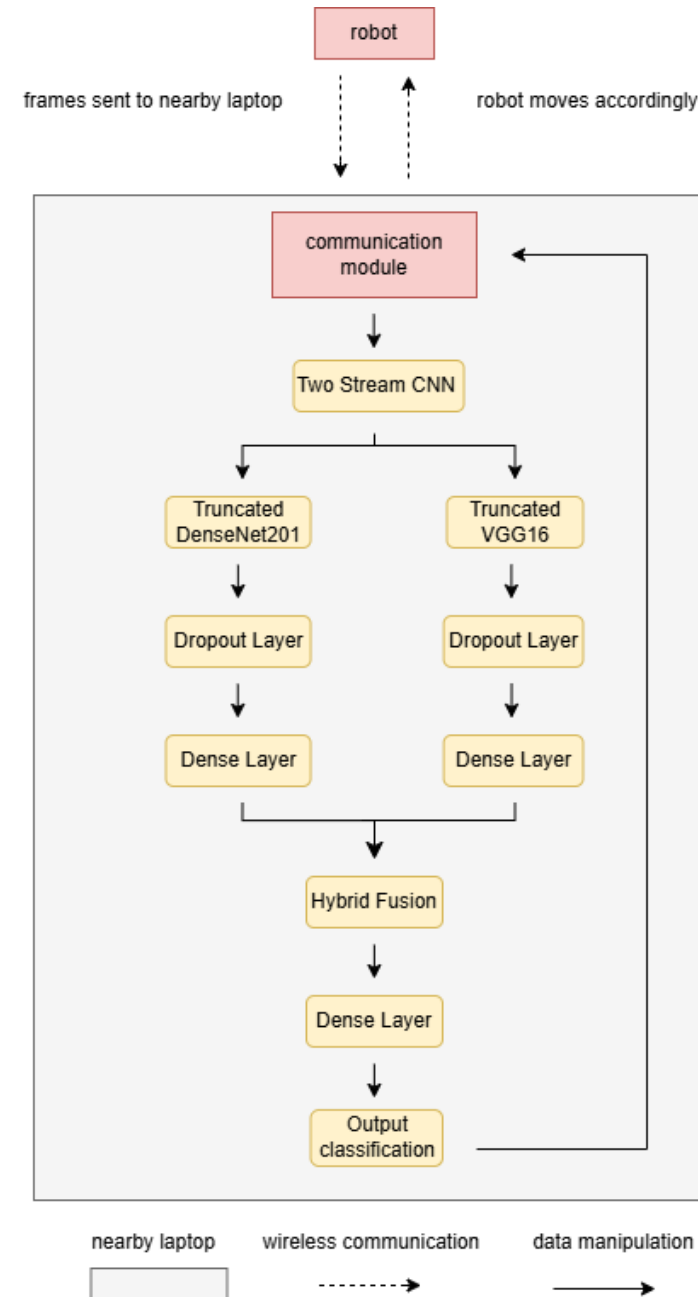
Data Architecture

- Laptop Robot Communication



Data Architecture

- Laptop Robot Communication
- Machine Learning Architecture using Pre-trained Models (training + validation)



Results



Fusion model improved cross-dataset performance: 78% on Kaggle (trained on My Images) 82% on My Images (trained on Kaggle)



Average prediction time: 0.24s



Average total response time: 2.63s



Fusion model = better generalization + real-time capable

Conclusion

- Demonstrated strong accuracy across environments
- Real-time response makes it suitable for accessibility tools
- Promising step toward practical, inclusive HMI systems